Sylwester Arabas | Curriculum Vitae | October 2026

contact

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public profiles

research group: open-atmos-krk.github.io
ORCID: orcid.org/0000-0003-2361-0082
GitHub: github.com/slayoo
Google Scholar: scholar.google.com/citations?user=X1s5grkAAAAJ
LinkedIn: linkedin.com/in/sylwester-arabas

highlights

· **numerical modelling** and data analysis in atmospheric physics
· cloud physics, aerosol-cloud-precipitation interactions, particle-based μ-physics
· **research software engineering**, focus on reproducibility & maintainability
· free and open-source software maintenance, dissemination and advocacy
· scientific **data visualisation**, vector graphics and typesetting
· public presentations, teaching sciences, organisation of meetings
· building teams, **keeping things simple, done and documented**
· international experience in academic, gov, corporate and startup realms

employment

2023.05–...: **AGH Univ. (Physics/Appl. CS), Kraków, Poland**
lecturer and researcher at the Environmental Physics Group: zfs.agh.edu.pl
2021.07–2022.06: **Univ. Illinois (Atmos. Sci.), Urbana, Illinois, USA**
postdoc at the group of Nicole Riemer: atmos.illinois.edu/~nriemer
2018.10–2023.04: **Jagiellonian Univ. (Computer Sci.), Kraków, Poland**
open-source Python project leadership: PySDM, PyMPDATA, Numba-MPI
2017.10–2018.09: **AETHON, Athens, Greece**
urban transport modelling (EU's H2020 "Innovation Associate" program)
2015.11–2017.09: **Chatham Financial, Kraków, Poland**
financial models software development
2014.01–2015.10: **University of Warsaw ([Geo]Physics), Poland**
open-source CFD C++ project leadership, lecturing C++ to 1st-year students
2005.10–2013.12: graduate studies (see below)
2002.10–2005.12: **Mazovian Governor Office, Warsaw, Poland**
backend web development
2000–2009: **ITStudio.pl, Warsaw, Poland**
backend web development

academic degrees

2026: **Jagiellonian University in Kraków** – Habilitation in Earth Sciences
degree obtained: 2026-07-10
2008–2013: **University of Warsaw** – PhD in Physics
thesis: Elements of modern cloud modelling (in English)
supervisor: H. Pawłowska, degree obtained: 2013-12-16
referees: G. Feingold (NOAA), L. Łobocki (Warsaw Tech.)
2002–2008: **University of Warsaw** – MSc in Physics, 350 ECTS
thesis: Microphysical properties of shallow convective clouds (in Polish)
supervisor: H. Pawłowska; referee: K. Haman; degree obtained: 2008-06-25

coding skills

Python, C++, C#, C, IDL/GDL, Fortran, SQL, UNIX tools
multi-threaded, GPU and MPI parallelism
design patterns; test and packaging automation
typesetting in LaTeX/BibTeX, Quarto, vector graphics

language skills

fluent: **Polish**, **English**;
conversational: Russian, French, Spanish; basics: Japanese

funding record

Poland's National Science Centre (ncn.gov.pl):
- 2022-26: PI in a SONATA project (ca. $250 000)
- 2011-13: PI in a PRELUDIUM project (ca. $15 000)

Foundation for Polish Science (fnp.org.pl):
- 2018-21: PI in "Reintegration" grant (ca. $200 000)
- 2014: Mentorship programme (mentor: prof. Harm Jonker, TU Delft)
- 2012-13: 2×START fellowship (incl. visit at NCAR, Boulder, CO)
- 2011: Conference award (SIAM GS11, Long Beach, California)

European Facility for Airborne Research (eufar.net):
- 2008: PI in SEASALT student project (seasalt.igf.fuw.edu.pl) (ca. $25 000)

open-source software

PyPartMC (2021–...): ∼ 500 C++ /Fortran/Python commits, maintenance
Numba-MPI (2020–...): ∼ 100 Python commits, maintenance
PyMPDATA & PySDM (2019–...): ∼ 3000 Python commits, maintenance
libmpdata++, libcloudph++ (2013–2015): ∼1000 C++ commits
GNU Data Language (2009–...): ∼500 C++ commits, co-maintenance

organisation of meetings

"Advances in Numerical Modeling of Aerosol-Cloud Interactions: ...":
- AMS Symposia on Cloud Physics & ACCI (joint), Houston, 2026:

"Probabilistic Particle-Based Methods in Aerosol-Cloud Microphysics":
- AMS Symposia on Cloud Physics & ACCI (joint), Baltimore, 2024
- AMS Symposium on Aerosol–Cloud–Climate Interactions, Denver, 2023

"Lagrangian cloud microphysics: progress and prospects" (EGU GA, 2020)
"Eulerian/Lagrangian methods for cloud microphysics":
- https://slayoo.github.io/workshop_2019/ (Kraków, 2019)
- https://goo.gl/1fj5H8 (Warsaw, 2015)

"FOSS for scientists" (Brussels, 2013, day-long conference session)

study visits

2025: (3 weeks) Johannes Gutenberg-Universität Mainz (Germany)
2022: (2 weeks) California Institute of Technology (Pasadena, California)
2015: (4 weeks) University of Hyogo (Kobe, Japan)
2012: (4 weeks) National Center for Atmospheric Research (Boulder, Colorado)
2010: (4 weeks) JAMSTEC/The Earth Simulator Center (Yokohama, Japan)

field campaigns

2011: (3 weeks) CARRIBA helicopter measurements campaign (Barbados)
2008: (3 weeks) EUCAARI aircraft measurements campaign (Rotterdam)
2008: (2 weeks) SEASALT aircraft measurements campaign (Austrian Alps)
2006: (3 weeks) AMMA aircraft measurements campaign (Burkina Faso)

extramural seminars

Freie Universität Berlin (2026)
University of Southern California, L.A. (2025)
Dept of Land, Air, and Water Resources, Univ. California, Davis (2025)
Institute for Atmopsheric Physics, Univ. Mainz (2025, 2022 virtual, 2019)
Columbia University / LEAP STC (2024): youtu.be/PgLGlfSPE7E
Environ. & Climate Sci. Dept, Brookhaven National Lab, NY (2022)
SoMAS, Stony Brook University, NY (2022)
Dept Environ. Sci. & Engineering, Caltech (2022): youtu.be/OOJe-JFMDpU
Dept Atmospheric Sciences, Univ. Illinois at Urbana-Champaign (2021)
Dept Atmospheric and Oceanic Sciences, McGill University, Montreal (2019)
National Center for Atmospheric Research, Boulder, Colorado (2019,'14,'12,'10)
Los Alamos National Laboratory (2019)
Faculty of Sciences, University of Pécs, Hungary (2019)
Department of Atmospheric Sciences, Yonsei University, Seoul (2019)
Graduate School for Simulation Studies, University of Hyogo, Kobe (2019,'15)

Nanjing University of Information Science and Technology, China (2019)
Lab. de mécanique des fluides et d'acoustique, École Centrale de Lyon (2019)
Physics Seminar, Michigan Tech, Houghton, Michigan (2018)
Dept of Atmospheric Sciences, University of Wyoming, Laramie (2018,'15)
Complex Systems and Applications Group, Demokritos, Athens, Greece (2018)
Chemical Engineering Department, University of Patras, Greece (2018)
Faculty of Civil Engineering and Geosciences, TU Delft, The Netherlands (2015)
National Atmospheric and Oceanic Administration, Boulder, Colorado (2012)
Meteorological Research Institute, Tsukuba, Japan, (2010)
Japan Agency for Marine-Earth Science and Technology, Yokohama (2010)

conference presentations

FOSDEM (Free & Open Source Software Devs Euro Meeting, Brussels):
'26, '25, '24 (PyPartMC talk), '23 (Numba-MPI talk),'21,'20,'19, '18,'17,'16,'15, '14,'13 (FOSS for Scientists devroom convener),'12,'11 (GDL talk),'10
International Conference on Clouds and Precipitation:
'24 (Jeju, poster), '21 (e-poster), '12 (Leipzig, talk), '08 (Cancún, talk)
American Meteorological Society Annual Meeting:
'24 (Baltimore, chair+talk), '23 (Denver, chair+poster), '22 (virtual, talk)
American Meteorological Society Cloud Physics Conference:
'22 (Madison, poster), '18 (Vancouver, poster)
UCAR Software Engineering Assembly Conference (Boulder, Colorado):
'21,'13 (talk)
European Geosciences Union General Assemblies (Vienna):
'20 (virtual, co-convener), '10 (poster), '09 (poster), '07 (poster)
Numerical Analysis and Scientific Computation with Applications:
'18 (Kalamata, talk)
C++ Now by Boost & Software Freedom Conservancy (Aspen, Colorado):
'15 (talk)
SIAM Conference on Mathematical and Computational Issues in Geosciences:
'13 (Padua, talk), '11 (Long Beach, talk)
American Geophysical Union Fall Meetings (San Francisco):
'12 (poster), '10 (poster)
Metström: Multiple Scales in Fluid Mechanics and Meteorology (Berlin):
'11 (talk)

workshops, schools, courses

2026: EuroSciPy (co-presenting a tutorial, AGH, Kraków)
2026: European Seminar on Computing ESCO (U. West Bohemia, Pilsen)
2025: Particle-based cloud modeling (U. Hyogo, Kobe)
2024: Dynamics of complex particles in turbulence (IRPHE, Marseille)
2023: Clouds containing ice particles (Univ. Mainz)
2023: Complex Dynamics of Particles in Turbulence (INRIA, Sophia Antipolis)
2022: 2-nd QuIESCENT Workshop / Arctic Science Summit Week (virtual)
2022: 4-th International workshop on Cloud Turbulence (NITech, virtual)
2021: Software Carpentry instructor training (virtual)
2021: 10-th International Cloud Modelling Workshop (virtual)
2021: Advanced numerical methods for hyperbolic equations (U. Trento)
2019: Water Isotopes and Climate (NCAR)
2019: DYAMOND-ESiWACE Hackathon (Mainz)
2013: ITM COMPLETE Workshop (Warsaw)
2018: Particle-based modeling of cloud microphysics (U. Hyogo, Kobe)
2018: Mathematics Applied in Transport and Traffic Systems (TUDelft)
2018: Innovation Management (A.T. Kearney, Dusseldorf/Berlin/Munich)

2017: Pedestrian Dynamics: Modelling, Validation and Calibr. (Brown Univ.)
2017: Robust Mathematical Finance (ETH)
2017: Quantitative Finance (U. Milano-Bicocca)
2016: Numerical methods for Hamilton-Jacobi equations (RICAM, Linz)
2014: IP, Licensing and Commercialisation (U. Oxford)
2014: Global Cloud Resolving Modelling (RIKEN, Kobe)
2014: Experim. Methodology in Comp. Sci. Research (U. St. Andrews)
2012: 8-th International Cloud Modelling Workshop (U. Warsaw)
2011: Atmospheric Water Vapour in the Climate System (Venice Int. Univ.)
2008: Aerosols and Climate Change (U. L'Aquila)
2008: Physics and chem. of air pollution and their effects (U. Helsinki)
2007: Boundary-Layer Research with Airborne Instruments (EUFAR, Iasi)
2007: Formation and growth of atmospheric aerosols (U. Helsinki)
2006: Multi-spectral environmental satellites (IMiGW/U. Wisconsin, Kraków)

student mentorship

PhD (co-advised with Mirosław Zimnoch):
– Agnieszka Żaba (PhD in Physics @agh.edu.pl, ongoing)
MSc:
– Paweł Magnuszewski (MSc in Applied CS @agh.edu.pl, defended in 2025)
– Oleksii Bulenok (MSc in CS @uj.edu.pl, defended in 2023)
– Kacper Derlatka (MSc in CS @uj.edu.pl, defended in 2023)
– Michael Olesik (MSc in physics @uj.edu.pl, defended in 2020)
– Piotr Bartman (MSc in CS @uj.edu.pl, defended in 2020)
BEng:
– Aleksandra Strząbała (BEng in Physics @agh.edu.pl, defended in 2026)
– Michał Wroński (BEng in Physics @agh.edu.pl, defended in 2026)
– Daria Klimaszewska (BEng in Physics @agh.edu.pl, defended in 2026)
– Michał Kowalczyk (BEng in Physics @agh.edu.pl, defended in 2025)
Internships:
– Emma Ware (Erasmus+ @agh.edu.pl, 2024/2025)
– Agnieszka Makulska (summer internship @agh.edu.pl in 2023)
– Zach D'Aquino (summer internship @illinois.edu in 2022)

university teaching

Dept Atmospheric and Climate Science, University of Washington Seattle:
2026: Hacking aerosol-cloud μ-physics modeling concepts in Python (3×3h short course, Erasmus+ staff mobility programme)
Faculty of Physics and Applied CS, AGH University of Krakow:
2026, '25 (lecture+lab), '24 (lab): Modelling of flow systems
2026, '25, '24: Fundamentals of Computer Science (lab)
2026, '25, '24, '23: Fundamentals of programming (C/Python/Julia lab)
2026 (lecture+lab): RSE in Python for Reproducible Comput. Science
2025: Modelling of physical systems (lab)
2024: Linux operating system (lab)
2024: C++ for technical physics (lab)
Faculty of Math. and CS, Jagiellonian University:
2020: Modelling of Atmospheric Clouds (lecture + computer lab)
2020: Programmer's Workshop (UNIX/LaTeX/git/...) (remote lab)
2020: Programming 1 (C/C++) (remote lab)
2018: Abstract programming (computer lab)
2018: Design patterns (computer lab)
Faculty of Physics, U. Warsaw:
2015: Programming in C++ (lecture)

U. Vigo in Ourense, Spain:
2014: A short course on object-oriented numerics (ephyslab.uvigo.es/numeric)

Institute of Geophysics, U. Warsaw:
2011, '14: Numerical modelling in atmospheric physics
2010: Physics of the atmospheric boundary layer
2009, '10: Atmospheric thermodynamics and cloud physics
2008, '09: Hands-on data processing in meteorology

peer-reviewed papers

Lüttmer, Arabas & Spichtinger 2026:
Stochastic homogeneous freezing of supercooled droplets in mixed-phase clouds in particle-based idealised microphysics framework
(GMD, doi: 10.5194/gmd-19-10185-2026)

Ware, Bartman-Szwarc, Igel & Arabas 2026:
Addressing the Collision-Coalescence Deficit in the Super-Droplet Monte Carlo Scheme with Adaptive Time-Stepping
(JAMES, doi: 10.1029/2026MS005773)

Tang, Arabas, Curtis, Knopf, West & Riemer 2026:
The impact of aerosol mixing state on immersion freezing: Insights from classical nucleation theory and particle-resolved simulations
(ACP, doi: 10.5194/acp-26-9221-2026)

Arabas, Curtis, Silber, Fridlind, Knopf, West & Riemer 2025:
Immersion freezing in particle-based aerosol-cloud microphysics: a probabilistic perspective on singular and time-dependent models
(JAMES, doi: 10.1029/2024MS004770)

Derlatka, Manna, Bulenok, Zwicker & Arabas 2024:
Numba-MPI v1.0: Enabling MPI communication within Numba/LLVM JIT-compiled Python code
(SoftwareX, doi: 10.1016/j.softx.2024.101897)

Azimi, Jaruga, de Jong, Arabas & Schneider 2023:
Training warm-rain bulk microphysics schemes using super-droplet simulations
(JAMES, doi: 10.1029/2023MS004028)

D'Aquino, Arabas, Curtis, Vaishnav, Riemer & West 2023:
PyPartMC: A Pythonic interface to a particle-resolved, Monte Carlo aerosol simulation framework
(SoftwareX, doi: 10.1016/j.softx.2023.101613)

de Jong, Mackay, Bulenok, Jaruga & Arabas, 2023:
Breakups are Complicated: An Efficient Representation of Collisional Breakup in the Superdroplet Method
(Geosci. Model Dev., doi: 10.5194/gmd-16-4193-2023)

de Jong et al., 2023
New developments in PySDM and PySDM-examples v2: collisional breakup, immersion freezing, dry aerosol initialization, and adaptive time-stepping
(J. Open Source Soft., doi: 10.21105/joss.04968)

Hill, Lebo, Andrejczuk, Arabas, Dziekan, Field, Gettelman, Hoffmann et al., 2023:
Toward a numerical benchmark for warm rain processes
(J. Atmos. Sci. doi: 10.1175/JAS-D-21-0275.1)

Park, Duvert, Coulais, Jung, Arabas et al. 2022:
GNU Data Language 1.0: a free/libre and open-source drop-in replacement for IDL/PV-WAVE
(J. Open Source Soft. doi: 10.21105/joss.04633)

Bartman et al. 2022:
PyMPDATA v1: Numba-accelerated implementation of MPDATA with examples in Python, Julia and Matlab
(J. Open Source Soft. doi: 10.21105/joss.03896)
Bartman et al. 2022:
PySDM v1: particle-based cloud modelling package for warm-rain microphysics and aqueous chemistry
(J. Open Source Soft. doi: 10.21105/joss.03219)
Olesik et al. 2022:
On numerical broadening of particle-size spectra: a condensational growth study using PyMPDATA 1.0
(Geosci. Model Dev. 15, doi: 10.5194/gmd-15-3879-2022)
Arabas & Farhat 2020:
Derivative Pricing as a Transport Problem: MPDATA solutions to Black-Scholes-type equations
(J. Comput. Appl. Math. 373, doi: 10.1016/j.cam.2019.05.023)
Arabas & Shima 2017:
On the CCN (de)activation nonlinearities
(Nonlin. Proc. Geophys. 24, doi: 10.5194/npg-24-535-2017)
Arabas, Jaruga, Pawlowska & Grabowski, 2015:
libcloudph++ 1.0: a single-moment bulk, double-moment bulk, and particle-based warm-rain microphysics library in C++
(Geosci. Model Dev. 8, doi: 10.5194/gmd-8-1677-2015)
Jaruga, Arabas, Jarecka, Pawlowska, Smolarkiewicz & Waruszewski, 2015:
libmpdata++ 1.0: a library of parallel MPDATA solvers for systems of generalised transport equations
(Geosci. Model Dev. 8, doi: 10.5194/gmd-8-1005-2015)
Arabas, Jarecka, Jaruga & Fijałkowski, 2014:
Formula Translation in Blitz++, NumPy and Modern Fortran: A Case Study of the Language Choice Tradeoffs
(Sci. Prog. 22, doi: 10.3233/SPR-140379)
Arabas & Shima, 2013:
Large-Eddy Simulations of Trade Wind Cumuli Using Particle-Based Microphysics with Monte Carlo Coalescence
(J. Atmos. Sci., doi: 10.1175/JAS-D-12-0295.1)
Kulmala et al., 2011:
General overview: European Integrated project on Aerosol Cloud Climate and Air Quality interactions (EUCAARI)
(Atmos. Chem. Phys., doi: 10.5194/acp-11-13061-2011)
Arabas & Pawlowska, 2011:
Adaptive method of lines for multi-component aerosol condensational growth and CCN activation
(Geosci. Model Dev., doi: 10.5194/gmd-4-15-2011)
Cairo et al., 2010:
An introduction to the SCOUT-AMMA stratospheric aircraft, balloons and sondes campaign in West Africa, August 2006: rationale and roadmap
(Atmos. Chem. Phys., doi: 10.5194/acp-10-2237-2010)
Arabas, Pawlowska & Grabowski, 2009:
Effective radius and droplet spectral width from in-situ aircraft observations in trade-wind cumuli during RICO
(Geophys. Res. Lett., doi: 10.1029/2009GL038257)

proposal reviews
- Fulbright Poland
- National Research, Development and Innovation Office of Hungary
- Dutch Research Council (NWO)
- German Research Foundation (DFG)

editorship

Geoscientific Model Development (EGU public-peer-review journal):
- 2020-:topical editor (geoscientific-model-development.net/editorial_board.html)
- 2021-24:"Particle-based methods for simulating atmospheric aerosol and clouds" special-issue coordinator (gmd.copernicus.org/articles/special_issue1164.html)

Jagiellonian University Press:
- -2023: Polish edition of "Short Course in Cloud Physics" by Rogers & Yau (project coordination, LaTeX typesetting) wuj.pl/ksiazka/zarys-fizyki-chmur